INCH-POUND

MS24166L
5 October 2020
SUPERSEDING
MS24166K
5 April 2018

DETAIL SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, 50 AMPERES, 1 PST,
(N.O.) TYPE II, NONHERMETICALLY SEALED

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall
consist of this specification and the latest issue of MIL-PRF-6106.

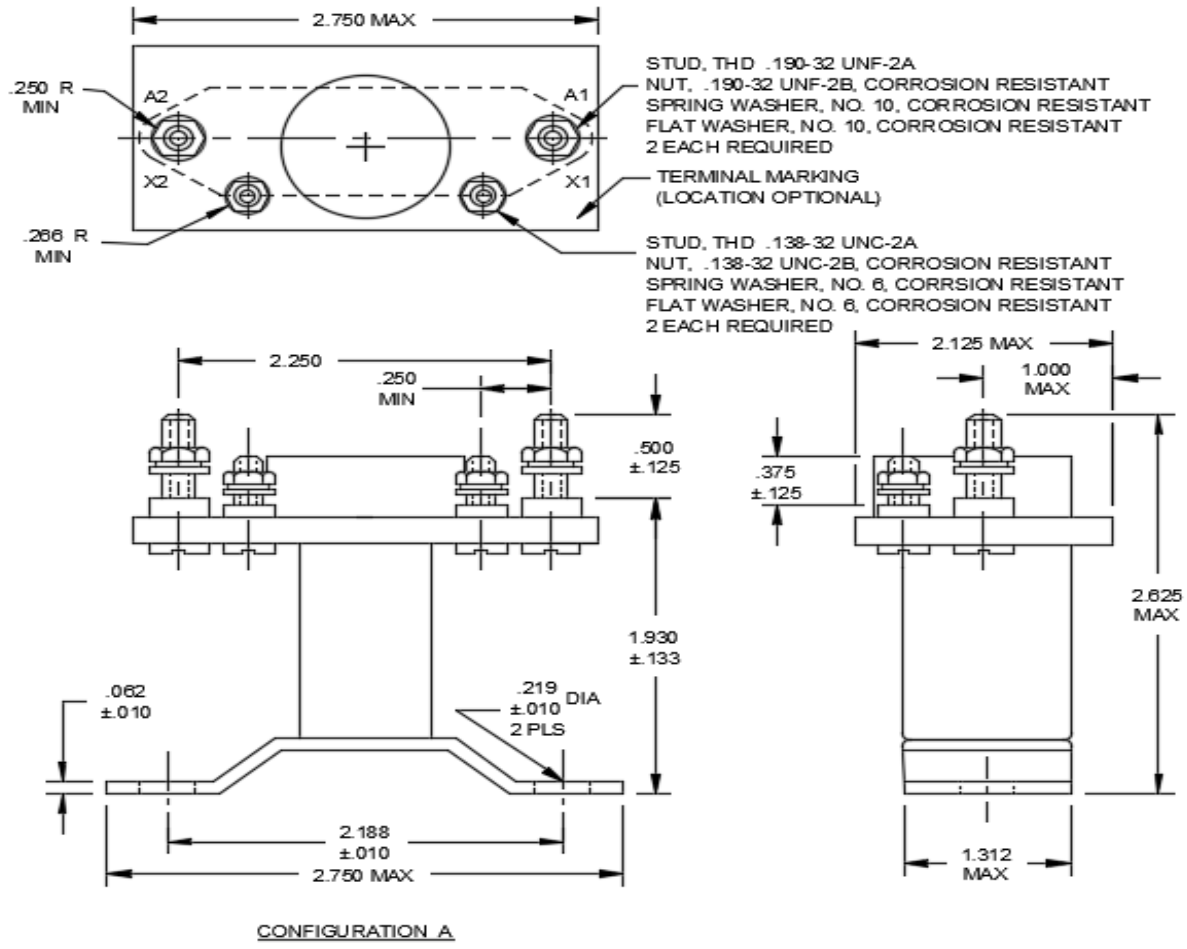
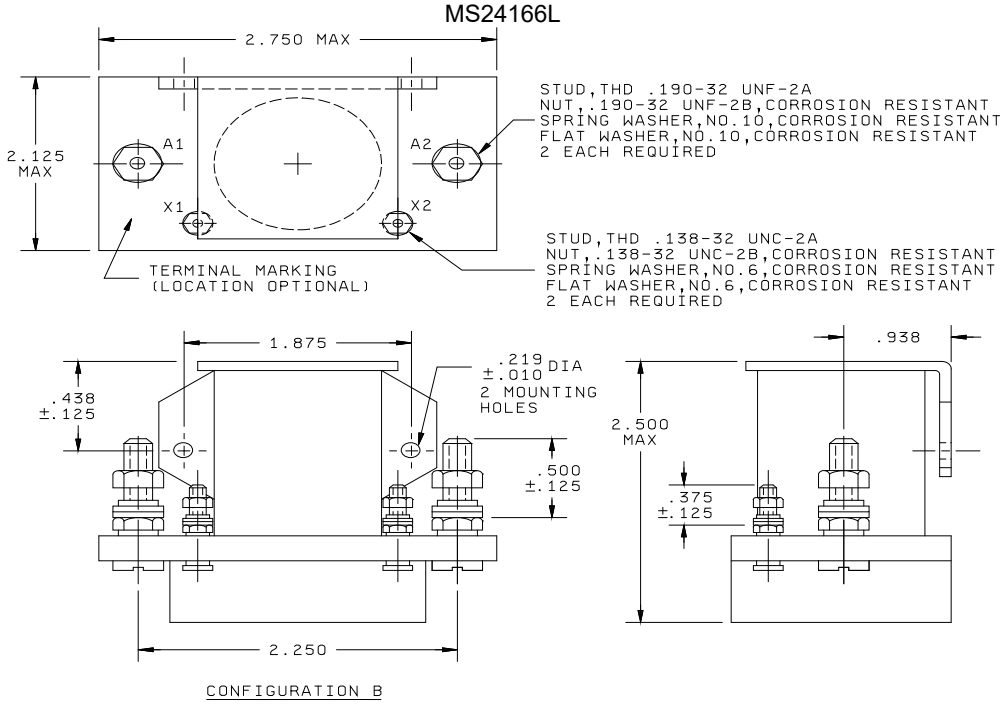


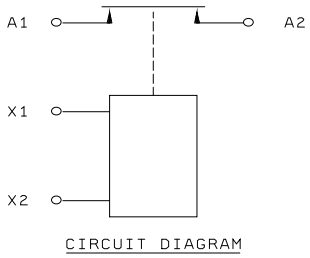

FIGURE 1. Configurations and dimensions.



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CONFIGURATION B




CIRCUIT DIAGRAM

| Inches | mm | Inches | mm |
|---|---|---|---|
| .010 | 0.25 | .938 | 23.83 |
| .062 | 1.57 | 1.000 | 25.40 |
| .093 | 2.36 | 1.312 | 33.32 |
| .125 | 3.18 | 1.875 | 47.63 |
| .133 | 3.38 | 1.930 | 49.02 |
| .219 | 5.56 | 2.125 | 53.98 |
| .250 | 6.35 | 2.188 | 55.58 |
| .266 | 6.76 | 2.250 | 57.15 |
| .375 | 9.53 | 2.625 | 66.68 |
| .438 | 11.13 | 2.750 | 69.85 |
| .500 | 12.70 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.031 (0.79 mm).
4. An additional flat washer may be used for terminal seat.
5. Terminal temperature rise under continuous current conditions is 95°C. Intermediate current shall be conducted at 71°C.
6. In the event of a conflict between the text of this specification and the references cited herein, this specification shall take precedence.
7. Referenced Government documents of the issue listed in that issue of the Acquisition Streamlining and Standardization Information System (ASSIST) specified in the solicitation forms a part of this standard to the extent specified herein.
8. Shape of relay is optional within the envelope dimensions shown.
9. Cadmium or cadmium compounds are prohibited on external hardware.
10. Spring washer on drawing is a spring lock washer.

FIGURE 1. Configurations and dimensions - Continued.



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REQUIREMENTS:

Configurations and dimensions: See figure 1.

Dash numbers and general characteristics: See table I.

Contact data:

Rated contact Load: See table II.

Maximum contact drop, initial: 0.150 V.

After life test: 0.175 V.

Overload current (NO): 400 amperes.

Rupture current (NO): 500 amperes.

Operating characteristics (see table III):

Duty rating: Continuous.

RFI specification: MIL-STD-461 (applicable to coil circuits of ac operated relays).

Electrical data:

Minimum insulation resistance:

Initial: 100 megohms.

After life or environmental test: 50 megohms.

Dielectric strength:

Sea level, 2-5 seconds:

| | Initial | | After life tests | |
|---|---|---|---|---|
| | 28 V dc | 115 V ac | 28 V dc | 115 V ac |
| Coil to case: | 1,250 V rms | N/A | 1,000 V rms | N/A |
| Aux. contacts: | 1,250 V rms | N/A | 1,000 V rms | N/A |
| All other points: | 1,250 V rms | N/A | 1,000 V rms | N/A |

Altitude, 1 minute.

| | Initial | |
|---|---|---|
| | 28 V dc | 115 V ac |
| Coil to case: | 500 V rms | N/A |
| Aux. contacts: | 500 V rms | N/A |
| All other points: | 500 V rms | N/A |



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Environmental characteristics:

Temperature range: -55°C to +71°C.

Maximum altitude rating: 50,000 feet.

Shock, g-level: 25 g's.

Duration: 6-9 ms.

Maximum duration contact opening: 2 ms.

Vibration levels, sinusoidal: See table IV.

Vibration, random: N/A.

High shock: N/A.

Acceleration: 10 g's.

Qualification by similarity: See MIL-PRF-6106.

Part or Identifying Number (PIN): MS24166- (plus applicable dash number from table I).

TABLE I. Dash numbers and general characteristics.

| PIN MS24166- | Type | Coil | Terminal type | Mounting or mating socket | Maximum weight (pounds) |
|---|---|---|---|---|---|
| D1 1/ | II | dc | Stud | Bracket - bottom | 0.6 |
| D2 2/ | II | dc | Stud | Bracket - side | 0.6 |

1/ For Government logistics support, MS24166-D1 shall be used in lieu of AN3350-2.
2/ For Government logistics support, MS24166-D2 shall be used in lieu of AN3350-1.



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TABLE II. Rated contact load (amperes per pole) case grounded.

| Type of load | Life operating cycles x $10^3$ | 28 V dc | | | | 115 V ac, 1 phase | | | | 115/200 V ac, 3 phase 1/ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz |
| Resistive | 50 | 50 | | | | | | | | | | | |
| Inductive | 10 | 50 | | | | | | | | | | | |
| Motor | 50 | 50 | | | | | | | | | | | |
| Lamp | 50 | 25 | | | | | | | | | | | |
| Transfer load 2/ | | | | | | | | | | | | | |
| Mechanical life (reduced current) | 100 | 12.5 | | | | | | | | | | | |
| Mixed loads | 50 | 5 | | | | | | | | | | | |

1/ Absence of value indicates that relay is not rated for 3 phase applications.
2/ Transfer load indicates that relay is suitable for transfer between unsynchronized ac power supplies at the rating indicated.

TABLE III. Operating characteristics.

| PIN MS24166- | Coil data | | | | | | | | | | | Time - (milliseconds maximum) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coil | Rated | | | Max | | Max pick-up voltage | | | Hold voltage 2/ | Drop-out voltage 2/ | Operate 3/ | Release 4/ | Bounce | | | |
| | | Volts 1/ | Freq. (Hz) | Res Ω +10% -15% | Volts | Ampere | Normal 2/ | High temp test | Cont Current test | | | | | Main | | Aux | |
| | | | | | | | | | | | | | | NO | NC | NO | NC |
| D1 | X1,X2 | 28 | dc | 94 | 29 | .35 | 18 | 21 | 22.5 | 7.0 | 1.5 | 20 | 10 | 5.0 | | | |
| D2 | X1,X2 | 28 | dc | 94 | 29 | .35 | 18 | 21 | 22.5 | 7.0 | 1.5 | 20 | 10 | 5.0 | | | |

1/ CAUTION: Use of any coil voltage less than rated coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With rated coil voltage.
4/ From rated coil voltage.

TABLE IV. Vibration levels (sinusoidal).

| 5-10 Hz | 10-55 Hz | 55-250 Hz | 250-500 Hz | 500-1500 Hz |
|---|---|---|---|---|
| 0.08 | .06 DA | 2 g's | 2 g's | N/A |



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The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.
Referenced documents.

In addition to MIL-PRF-6106, this document references the following:
MIL-STD-461

Custodian activities:
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5945-2020-052)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.



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